

Qatar-9b

R-2266

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-9_260412 · created 2026-08-16T01:49:45+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2461142.894 +/- 0.013 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.12 +/- 0.051
Transit depth (Rp/Rs)^2	1.43 +/- 1.21 [%]
Orbital Inclination (inc)	87.23 +/- 1.66
Airmass coefficient 1 (a1)	1.302 +/- 0.047
Airmass coefficient 2 (a2)	-0.113 +/- 0.017
Scatter in the residuals of the lightcurve fit is	3.54 %
Best Comparison Star	None
Optimal Aperture	2.1
Optimal Annulus	8.68
Transit Duration (day)	0.0654 +/- 0.0097

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.12 $\pm$ 0.051 vs 0.1489 (deviation 0.57 σ)
Duration fit vs published	0.0654 d vs 0.0779 d (deviation 1.29 σ)
Mid-time O-C	-13.9 min
Depth SNR	2.4
Residual scatter	3.54 %

Light curve

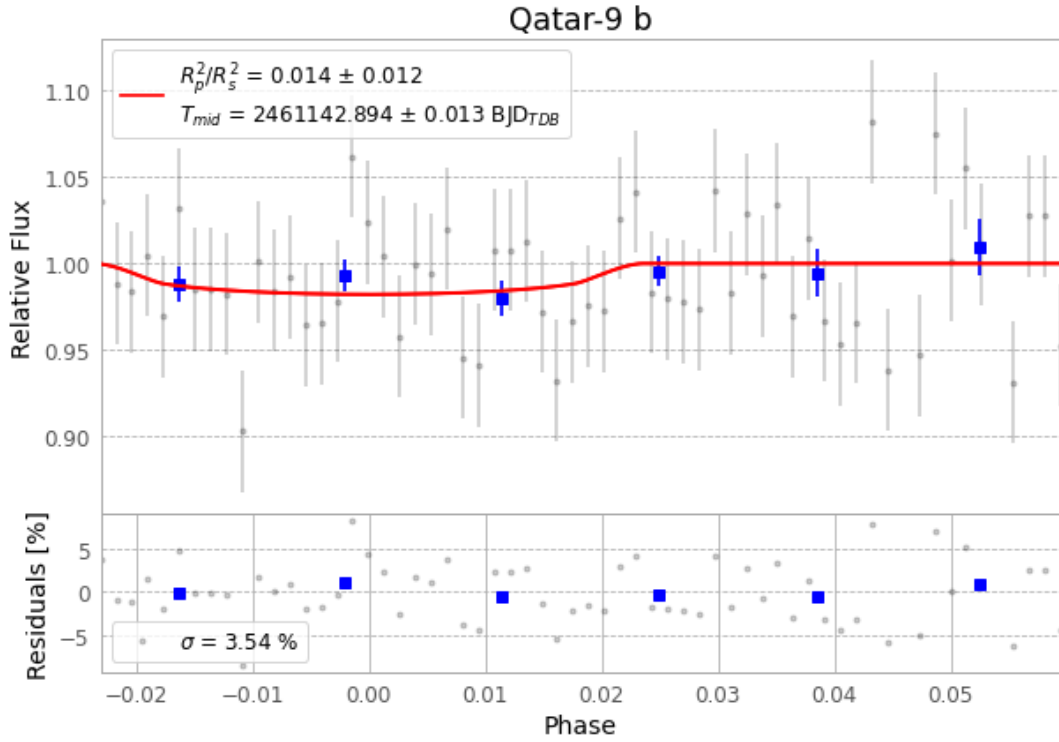


Figure 1 — FinalLightCurve_Qatar-9 b_2026-04-12.png (SHA-256 ce4ced3759df61a2...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [352, 206], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 0811e301d1633295...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

62 FITS frames (41 MB), Qatar-9260412083317.FITS → Qatar-9260412113615.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-9-260412.log (820f75608ad9...), FinalLightCurve_Qatar-9 b_2026-04-12.png (ce4ced3759df...), FinalLightCurve_Qatar-9 b_2026-04-12.csv (1662e4253f40...)

Compute resources

Wall time 97.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-16 01:49Z.